

SDEV2501

Lesson 15 – Discrete Visualization

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Today’s objective

- Create and interpret visualizations best suited for discrete data

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Agenda

- Discuss common discrete distributions
- Showcase types of visualizations well-suited for discrete data

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REVIEW

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Review

- What are examples of continuous distributions?
- What are some types of visualizations best used for continuous data?

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DISCRETE VISUALIZATION

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Discrete Variables

- Discrete data consist of **distinct**, **countable** values.
- Discrete numerical data often represents **counts**:
  - e.g. number of orders, software bugs, website visits, customer complaints

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Common discrete distributions: Binomial

- Models the number of successes in a **fixed number of independent trials**.
- Each trial has **two possible outcomes**, such as success/failure.
- The **probability** of success is **constant** across trials.

For example:

- How many heads do we get if we flip a fair coin 100 times?
- Of 20 software tests, how many pass if each test has a 90% probability of passing?
- Of 100 visitors, how many customers click a specific link?
- Of a sample of 25 products, how many items are defective?
- Of a fixed class size, how many students complete an optional activity?

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Common discrete distributions: Poisson

- Models **counts of events** within a defined **interval** of time, space, or another measure.
  - e.g. calls arriving per minute, bugs reported per day
- Often used when events occur **independently** at a stable **average** rate.
- The rate parameter λ (lambda) represents the **expected** count per interval.

For example:

- A help desk receives an average of four support tickets per hour. How likely is it to receive zero, four, or eight tickets in a particular hour?

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Limitations of theoretical distributions

- Real-world data can be messy:
  - Overdispersion (i.e. higher variability than expected)
  - Clustering
  - Seasonality
  - Excess zeros

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Expected Values

- The **expected value** is the long-run **average value predicted** by a probability distribution.
  - For a discrete random variable, multiply each possible value by its probability and add the results.
- Expected value **does not necessarily represent** the most likely outcome.
  - e.g. the expected value of a win when purchasing a lottery ticket
- An expected value can be a decimal even when individual outcomes must be whole numbers.
  - e.g. the expected value when a fair 6-sided die is rolled

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What's the expected number of tickets/hour?

If there is a

- 10% chance of 0 tickets/hour
- 20% chance of 1 ticket/hour
- 40% chance of 2 tickets/hour
- 20% chance of 3 tickets/hour
- 10% chance of 4 tickets/hour

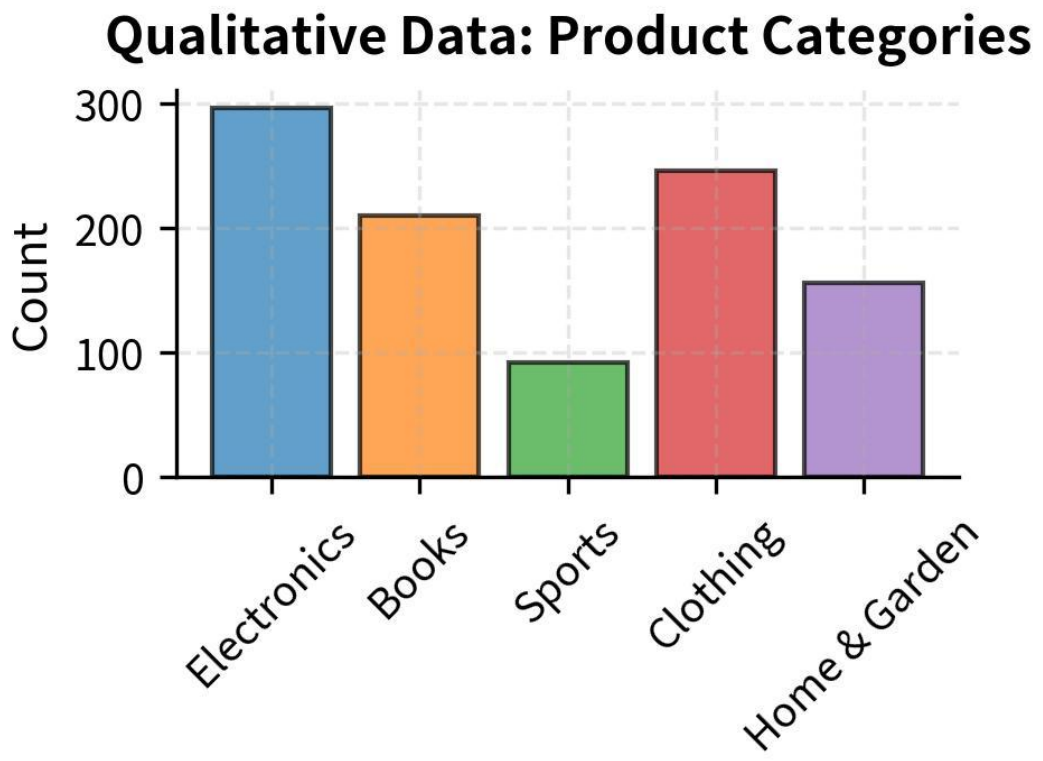
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WHAT TYPES OF VIZ ARE BEST FOR DISCRETE DATA?

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BAR CHART

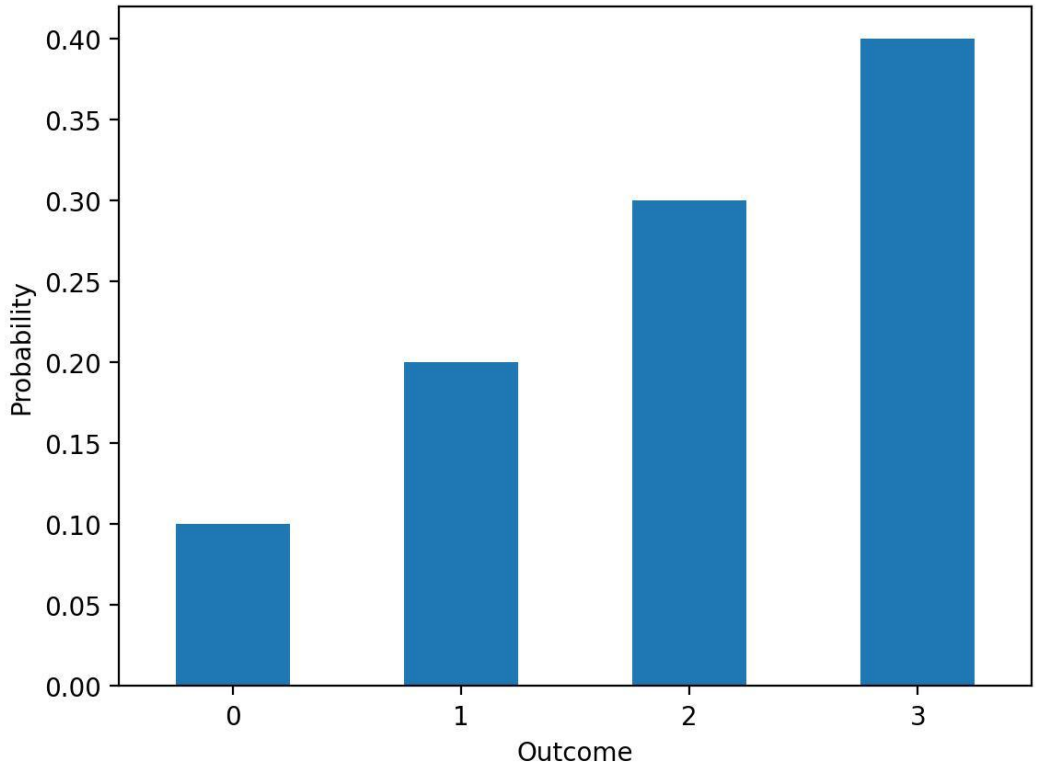
- **Compares** frequencies across categories or distinct values.
- Examples: tickets by priority, orders by product type, or number of bugs per project.
- Bars are separated to indicate distinct categories.
- Sort categories when ranking matters; retain a meaningful order for **ordinal** categories.



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DISCRETE PROBABILITY DISTRIBUTIONS

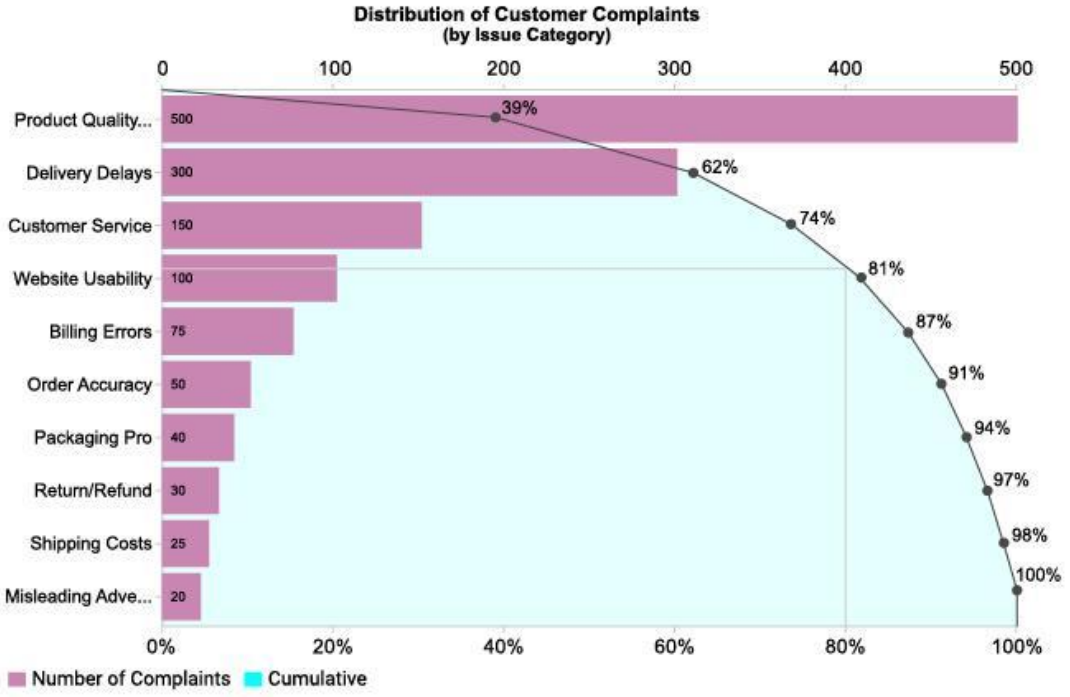
- Displays the **probability** of each **possible discrete outcome**.
- Examples: the number of defects in a batch or the number of arrivals in an hour.
- The probabilities across all possible outcomes sum to 1.
- Useful for **comparing observed** counts with a **theoretical** binomial or Poisson model.



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PARETO CHART

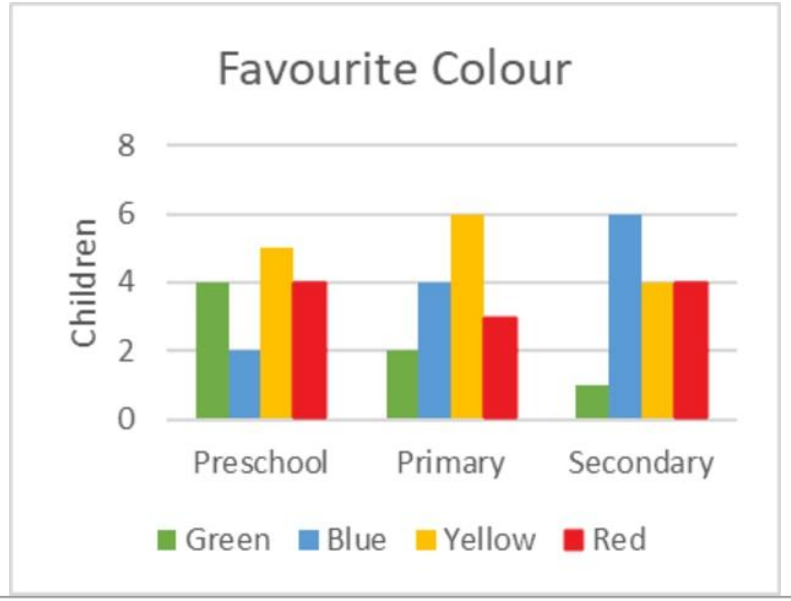
- **Ranks** categories from highest to lowest frequency.
- Often adds a cumulative-percentage line.
- Useful for identifying which complaint types, defects, or error categories contribute most to the total.
- A Pareto chart highlights concentration; it doesn't prove the causes of a problem.

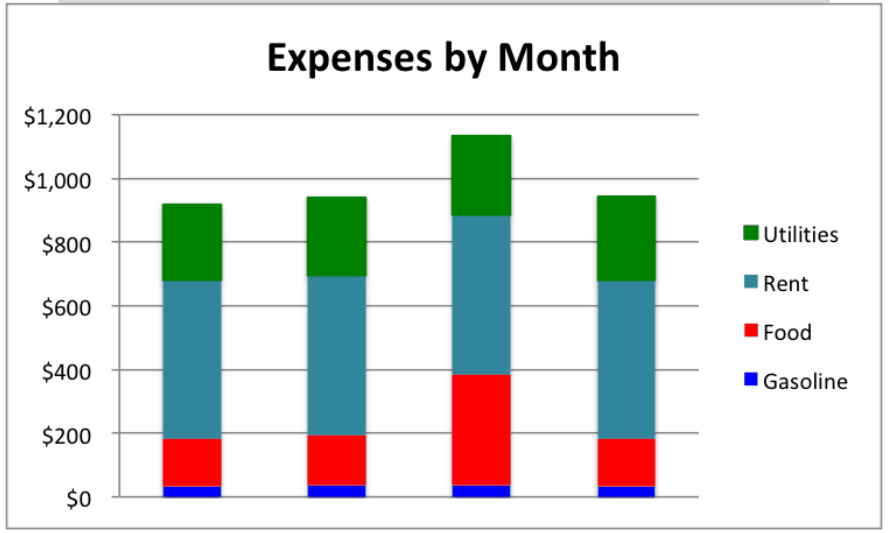


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GROUPED OR STACKED BAR CHART

- **Grouped** bars make it easier to **compare** counts across **groups**.
- **Stacked** bars show how a **total** is **composed** of several categories.
  - 100% stacked bars compare **proportions** rather than absolute **counts**.
- Always check group sizes: larger groups can have higher counts without having higher rates.





Counts vs Proportions

- **Counts** tell us how many observations fall into each category.
- **Proportions** or **percentages** tell us what share of a group falls into each category.
- Counts can be misleading when groups have different sizes: always identify the **denominator** when reporting a rate or percentage.
- e.g. Team A has 40 high-priority tickets out of 200 total tickets (20%). Team B has 25 high-priority tickets out of 50 total tickets (50%).
  - Team A has **more** high-priority **tickets** in absolute terms, but Team B has a **greater proportion** of high-priority tickets.
  - Which team has the bigger problem depends on the question and context.

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Other options

- **Dot plot**: Useful for displaying frequencies of a small number of discrete numerical values.
- **Mosaic plot:** Shows the relationship between two categorical variables using areas.
- **Heatmap**: Useful for showing counts across two categorical dimensions, such as weekday and ticket priority.
- **Line chart**: Appropriate for counts observed over ordered time intervals, such as tickets per day.

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Small Group Activity: Chart Choice Challenge

Choose a viz to answer each question:

1. Which product **category** has the **most orders**?
2. Which **day** had the **most support tickets**?
3. What **proportion of tickets** are **high priority**?
4. Which **complaint types** account for **most complaints**?
5. How does the **mix of ticket priorities** differ between **teams**?

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Python demo

- Load + summarize dataset
- Build a frequency bar chart
- Create a grouped or stacked bar chart
- Apply viz best practices
  - Color, descriptive labels, check units, etc

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Closing

- 1-minute reflection:
  - Can we answer the same types of questions with discrete vs continuous data?

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What’s next?

- Lying with visualization

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